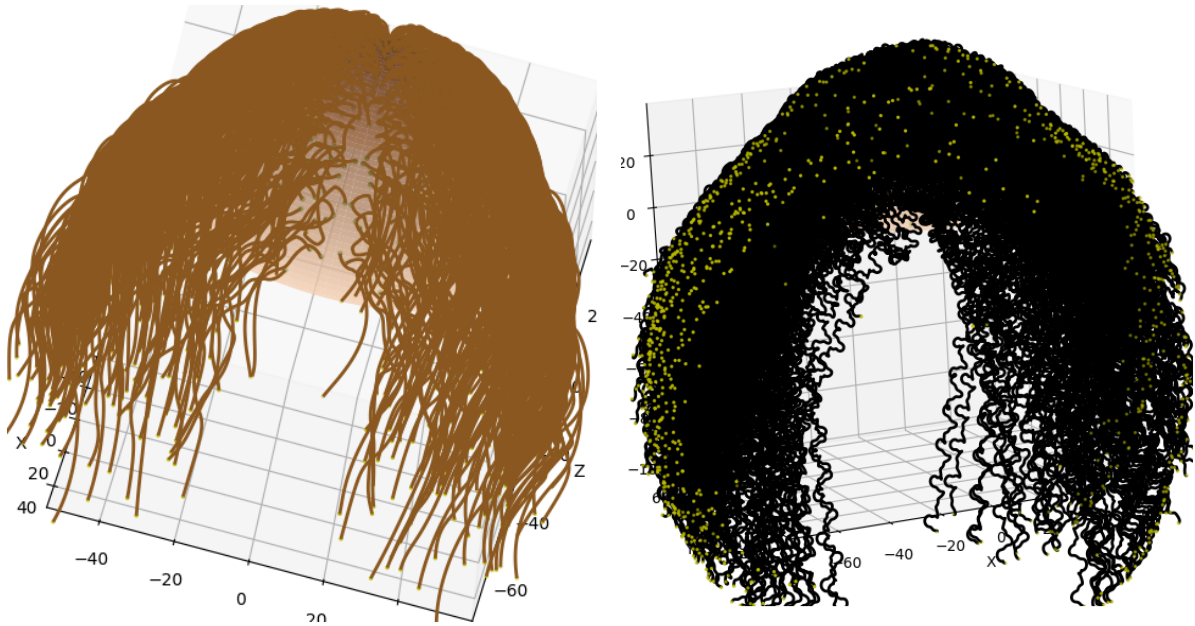


Hair Project Phase 0: Parting Line

by Amare McDonald



An initial Geometric Hair Model

First published: September 18, 2026

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Plagiarism declaration

This piece of work is a result of my own work. Material from the work of others not involved in the project has been acknowledged, quotations and paraphrases suitably indicated, and all uses of AI tools have been declared.

AI Usage Declaration

- AI was used to generate a working copy of (Bridson 2007)'s updated algorithm for random, Poisson Disk sample generation within a rectangular grid, as well as colour variation options. All other code and work in this report is my own.

Links

You're welcome to share this PDF with other professionals, partners and students — just keep it unchanged and credit me. If you would like more information about the code or process of construction, see my Github, otherwise feel free to contact me,

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Github: <https://github.com/DownFall454/Hair-Project-.git>

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1 Hair and Curves

1.1 Understanding Hair

Hair threads are natural fibres formed from keratin; their shaft structure is determined by processes during hair growth in the follicle, alongside spatiotemporally distributed ones of keratin maturation and cornification (Richena and Harland 2022). Curliness in hair is an emergent phenomenon, arising from internal organization of the hair cortex, follicle morphology, shape of the fibre cross-section, as well as differences in protein abundance. Many advances in the genomic causes of hair variation have been done with immunohistochemical approaches; few have a targeted understanding of curl. (Richena and Harland 2022)

The main physical properties of hair, elasticity, volume, smoothness etc. depend mostly upon its geometry. (Velasco et al. 2009) From an engineering perspective, a curved structure presents various physical and mechanical challenges when being manipulated; the distinct biomechanical characteristics of strands with high curvature make it prone to dryness and then breakage. (Cloete et al. 2019) Curly fibres have a behaviour reminiscent of a coiled spring; however, they betray a distinct toe-region behaviour before heeling into elastic deformation because of their viscoelastic nature. (Cloete et al. 2020) Hand manipulation of the fibre assembly imposes principally torsion and bending stresses on a single fibre, shearing between cortical cells due to torsional deformation. (Kamath et al. 1984) After much investigation, around a specific curvature, under the gravitational force, the fibre shifts from axial elongation to a kind of rotation in position.

Until recently, hair shape was classified using race-based taxonomies; Mettrie and Lousouarn typified hair into an 8-group classification scheme, for which the high curled category was found to contain a disproportionate amount of variation. Mkentane later collapses the latter into a 6-curl group for easier implementation. (Mettrie et al. 2007)

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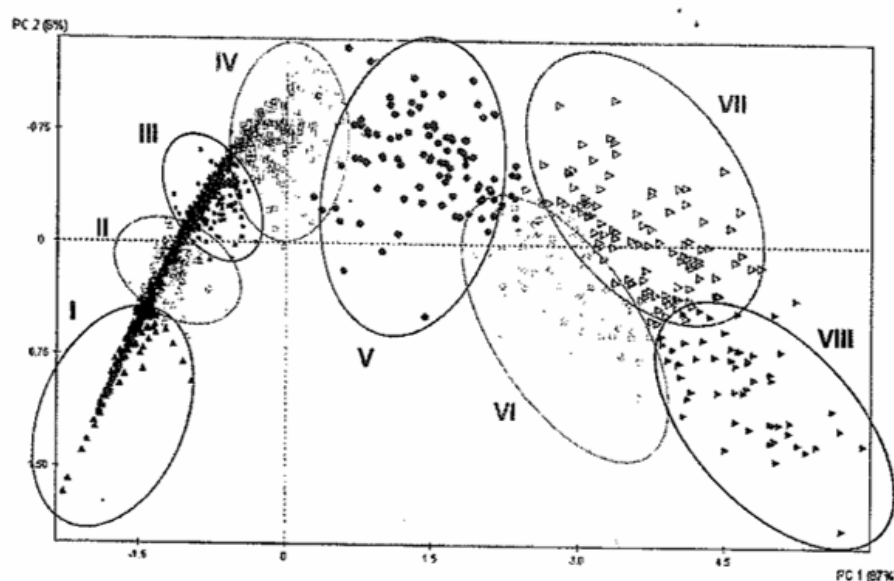


Figure 2: Principle component analysis with 8 cluster Partition

The first component PC1 covers around 87% of the total variation (Mettrie et al. 2007). Curve diameter (CD) is a key parameter for describing groups I – IV, after which curl index, alongside the number of waves and twists, the torsional aspects take over as the main cause of variation. Nevertheless, very few studies on the quantification of hair fibre geometry have reported according to hair curl type classification schemes; attributes that are true for the phenotype are often credited to the genotype. (Berg et al. 2025) (Cloete et al. 2019) The most popular categorization is the Walker typing system, 1A-4C, which is criticized for its ultimately qualitative measures. (Idowu et al. 2024)

(Mettrie et al. 2007) has the boundary of the latter four categories marked a curvature radius $< 12mm$, and therefore an implied curvature $> 0.0833mm^{-1}$. On the other hand, (Berg et al. 2025) marked it as $< 6mm$, and implied curvature $> 0.1667mm^{-1}$. However there is an extreme example from Berg et Al, a curvature diameter $0.2mm$ which implies maximum mathematical curvature of $1mm^{-1}$, which means that approximately 83% of hair curvature variation lies beyond this value, but this variation is not distributed equally.

The mechanical stiffening of keratin networks during hair formation follows a power law, moreover one that is “typical of the mechanics of nonthermal bending-dominated fibre networks”. (Bornschrögl et al. 2016) The probability distribution of curvature becomes localized around 0 over time (Du et al. 2021), and in biological materials has been constructed with an exponential tail. (Weber et al. 2015) Hair curvature is a continuous phenomenon. Under a specific mode of perception, the continuity of human variation is discretized, not only magnifying discontinuity, but magnifying variation in one box while minimizing variation in another. (Cloete et al. 2019) describes a gap in comprehensive understanding and a lack of interconnectedness in the study of hair variation. On the other hand, there is an asymmetric distribution of torsion in nature; in living systems there is a hierarchical bias towards right-handed chirality but is nevertheless intricately related to and follows a similar distribution to curvature. (Lininger et al. 2023)

The high incidence of fibre curvature points to the ‘ordinariness’ that curly fibres should enjoy when studying hair (Cloete et al. 2019). Curvature and torsion are caused by the chemistry and structural organization of the shaft (Richena and Harland 2022), which are informed by its geometry (Velasco et al. 2009). Phase 0 is therefore a purely geometric attempt at a Hair model that takes the range of human hair curvature as its starting point, up to a perception value k_p , which currently is fixed at 1. This phase does not explore the torsion domain, nor accommodate the cross-sectional areas of the strands, but models them as curves according to the Frenet Serret Equations.

1.2 Frenet Serret

A curve in $\mathbb{R}^3$ parameterized by arc length (p.b.a.l) can be defined by a function, $\alpha : I \rightarrow \mathbb{R}^3$, $I \subseteq \mathbb{R}$, and the Frenet Serret equations,

$$\begin{aligned} T'(s) &= \kappa(s)N(s) \\ N'(s) &= -\kappa(s)T(s) + \tau(s)B(s) \\ B'(s) &= -\tau(s)N(s) \end{aligned}$$

where $\alpha'(s) = T(s)$ is the unit tangent vector to the curve, and $N(s), B(s)$ the unit normal and binormal vectors under a right-handed frame, such that $B = T \times N$.

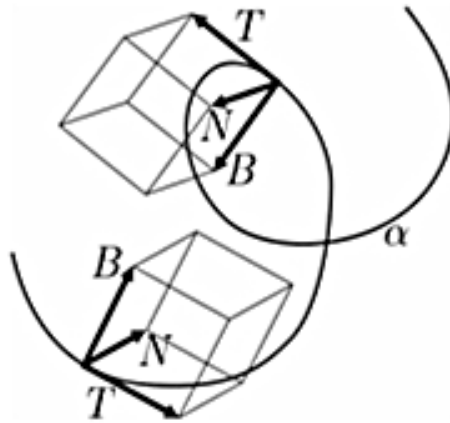


Figure 1.7. *Frenet trihedron*

We can express the Frenet Serret equations using the following.
let $v : I \rightarrow \mathbb{R}^3$, where

$$v(s) = \begin{pmatrix} T' \\ N' \\ B' \end{pmatrix} = \begin{pmatrix} 0 & \kappa(s) & 0 \\ -\kappa(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}$$

Which reduces the problem to a vector valued ordinary differential equation. Also note that (T, N, B) is an orthonormal basis, for which the curve can be expressed as a linear combination.

$$\alpha(s) = u(s)T(s) + v(s)N(s) + w(s)B(s) \quad u, v, w \in \mathbf{R}, s \in I$$

Upon differentiating and substituting the Frenet Serret equations

$$\alpha' = T = u'T + T'u + v'N + vN' + w'B + wB' \quad (1)$$

$$T = u'T + u(\kappa N) + v'N + v(-\kappa T + \tau B) + w'B + w(-\tau N) \quad (2)$$

$$= u'T + \kappa uN + v'N - \kappa vT + \tau vB + w'B - \tau wN \quad (3)$$

Then, recognizing (T, N, B) as an orthonormal basis, we can derive the following,

$$1 = u' - \kappa v$$

$$0 = \kappa u + v' - \tau w$$

$$0 = \tau v + w'$$

Upon rearrangement, we can find the similar form,

$$\begin{pmatrix} u' \\ v' \\ w' \end{pmatrix} = \begin{pmatrix} 0 & \kappa(s) & 0 \\ -\kappa(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

This was what I initially used when generating curves, being mathematically beautiful.

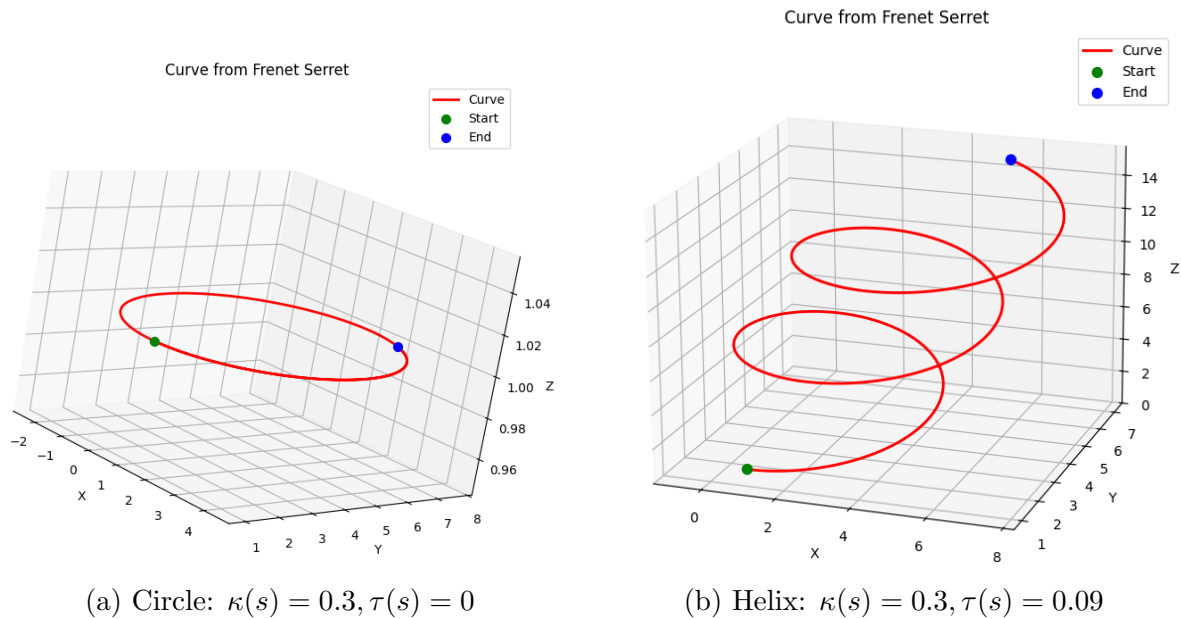
However later on in the project, solving for the coefficients and then multiplying throughout accumulated a substantial amount of numerical error, especially due to that forcing term w.r.t u ; it is much simpler to recognize that initial relationship,

$$\alpha'(s) = T(s)$$

So the position can be obtained solely from considering the tangent vector solution obtained from the Frenet frame. This is a basic ODE which can be solved through a variety of methods.

1.3 Curve Generation

Individual curves are uniquely determined by their curvature and torsion functions, $\kappa(s), \tau(s)$, over the interval I . For example, Constant curvature and 0 torsion defines a circle...



...while constant curvature and constant torsion defines a circular helix. More interesting things begin to occur when we let these properties vary continuously over the interval, such as

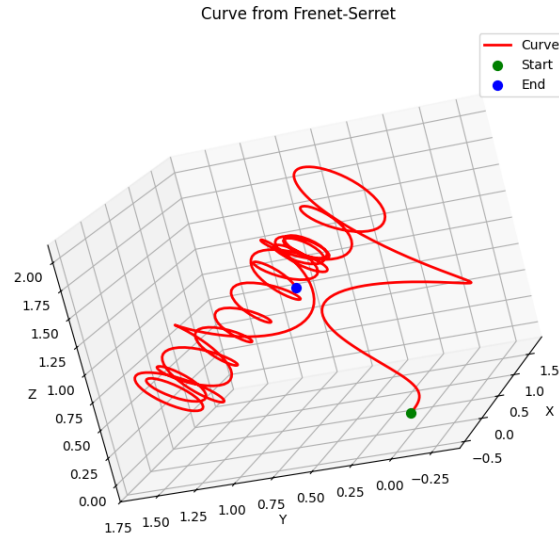


Figure 4: $\kappa(s) = \sin(s) + \sqrt{s}$, $\tau(s) = \cos(0.3s)$

However my attempts to mirror real world phenomena not including the torsion domain limited any extensive usage like the latter. There is more to consider when it comes to the specific functions chosen for curvature and its relationship length, which can be far from arbitrary. The torsion function, was fixed such that,

$$\tau(s) = \tau_{init} + A_{\tau} \sin(\kappa_{ave} \kappa_{init} s)$$

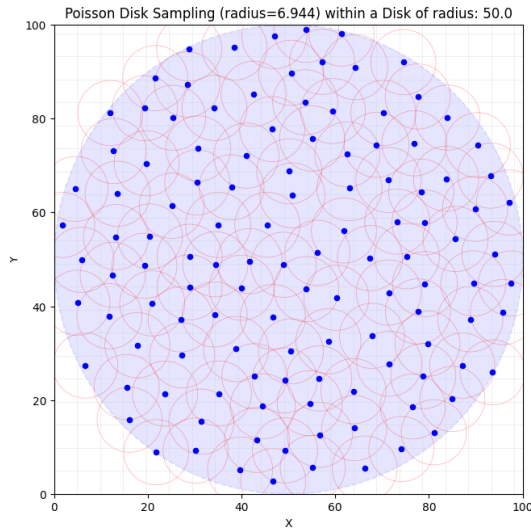
Where $\kappa_{init}/\tau_{init}$ are the initial values of that specific strand. In the distribution section

I will touch upon the curvature and torsion intervals, $[\kappa_1, \kappa_2], [\tau_1, \tau_2]$, with A_{τ} being an arbitrary combination of their averages. Having completed the curve generating aspect, I then moved onto the larger spatial frame where I could start approximating something that actually looks like a head.

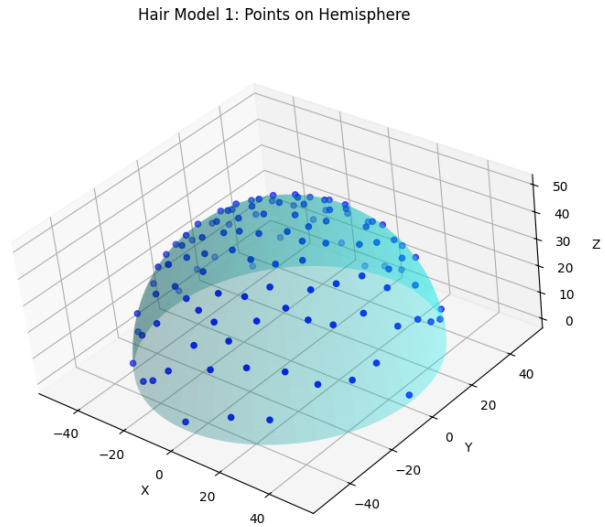
2 Distribution

2.1 Poisson Disk and Basic Shape

(Bridson 2007) presents an updated algorithm for random, Poisson Disk sample generation within a rectangular grid. With some minor alteration to conditions, and approximating a constant, one can rework it to generate a specific number of samples within a circle of radius R . I translated and then projected this circle of points onto a hemisphere of radius R centered at the origin.



(a) Poisson Disk Sampling, 121 points, $R = 50$



(b) Points on Hemisphere

2.2 Value assignment

Recall that the biomechanical processes during hair formation follow a power law, and the probability distribution of curvature has been seen to follow an exponential distribution. I chose to model curvature and torsion with the Pareto distribution, often used to model wealth disparities.

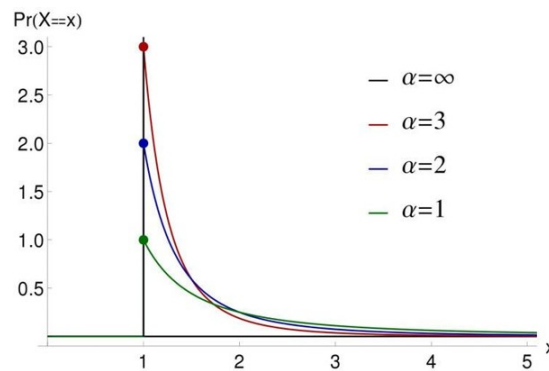


Figure 6: Pareto Distribution

For a random variable $X \sim \text{Pareto}(x_m, \alpha)$, where $x_m > 0$ is the scale parameter and $\alpha > 0$ is the shape parameter, its probability density function is given by,

$$f_X(x) = \begin{cases} \frac{\alpha x_m^\alpha}{x^{\alpha+1}} & \text{if } x \geq x_m \\ 0 & \text{if } x < x_m \end{cases}$$

I used inverse transform sampling to obtain a fixed number of pareto distributed curvature and torsion values ($\kappa_{init}, \tau_{init}$) within an interval of my choosing, with $1 \leq \alpha \leq 2$ for finite mean, and store them in a single vector. These will be the general values across the scalp (hemisphere), but may vary along the shaft, as you will see later. However, these vectors were not assigned to every root (point) across the scalp. Rather,

I assigned them to a given number of guide hairs; every other hair's assigned vector would then be a weighted average of their nearest guide hairs. To do this, the weights needed to satisfy a few conditions;

1. $0 < w_i \leq 1$, where i is the guide hair index
2. $\sum_i w_i = 1$
3. $w_i = f(d_i)$, where d_i is the distance to the nearest guide hair
4. f is a decreasing function, so the closer the guide hair, the greater weight it has

Having worked with it prior, the Pareto distribution emerged as beautifully applicable.

Let $D \sim \text{Pareto}(r_{min}, \alpha)$ be the distance to the nearest guide hair, where r_{min} is the minimum radius from the Poisson Disk sampling, and the shape parameter is identical to the one used in the curvature distribution (not torsion). Then,

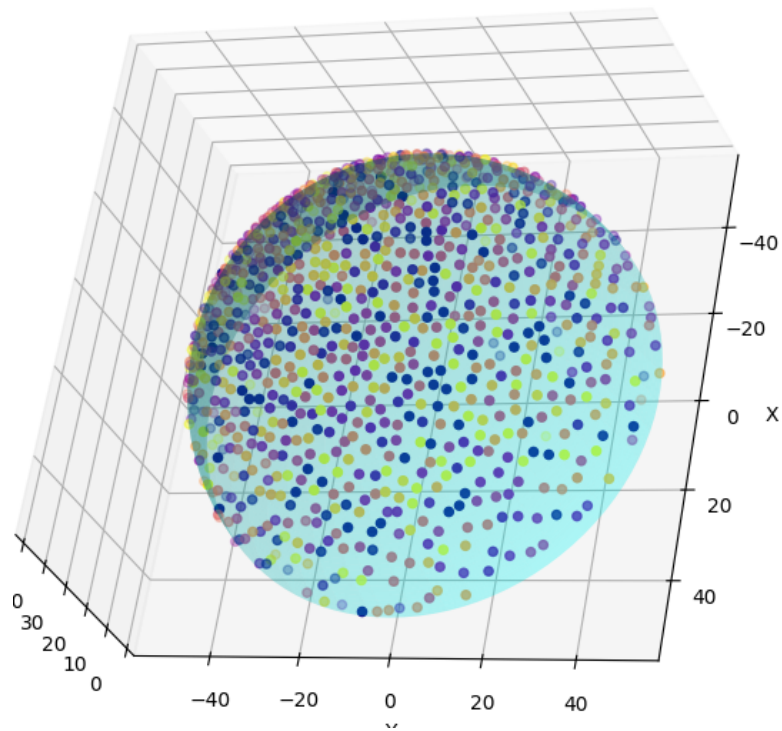
$$f_D(d) = \frac{\alpha r_{min}^\alpha}{d^{\alpha+1}} = w_i^*$$

But the weights need to satisfy the second condition, so

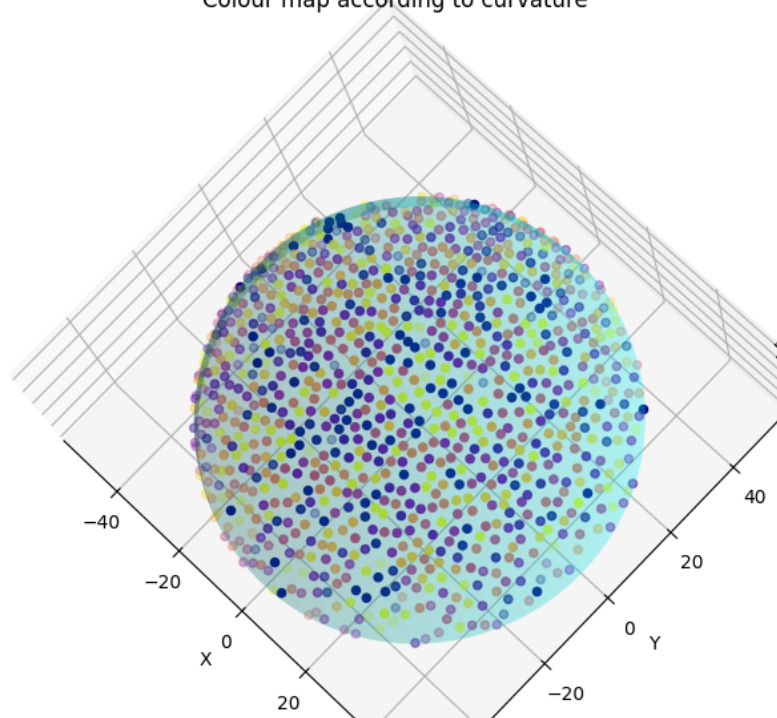
$$w_i = \frac{w_i^*}{\sum_i w_i^*}$$

Following Pareto, the curvatures are weighted towards the lower end of their intervals (blue). The chosen map colored according to their assigned curvature values can be seen below. Guide hairs were assigned curvature values, and then mirrored across the parting line and assigned the same curvature values, to ensure even distribution on both sides.

Colour map according to curvature



Colour map according to curvature



3 Direction

3.1 Tangent Plane

Now Direction was certainly the longest part of the project, determining not just where I wanted the hairs to go, but how I wanted this to vary across the scalp. My immediate focus was on the tangent plane to the surface; I wanted the shaft to be oriented normal to the scalp at every root. The normal vector to the tangent plane on the hemisphere at a point P is given by

$$e_3 = \frac{P}{\|P\|}$$

We can construct the vectors that span the tangent plane using the following,

$$v_1 = e_3 \times \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad v_2 = \hat{v}_1 \times e_3$$

Then,

$$e_1 = \hat{v}_2, \quad e_2 = e_3 \times e_1$$

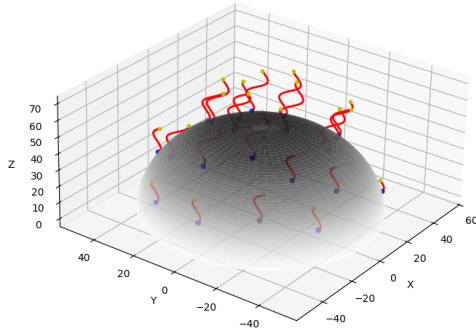
The latter for which the order is important, as recall these vectors can then be input into the solver as the initial conditions for the Frenet Trihedron, yet

$$\alpha'(s) = T(s)$$

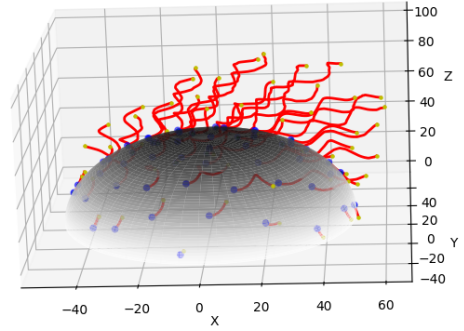
Namely, the curve initially moves in the direction of the tangent vector. So the tangential component of the Frenet frame must be normal to the surface at a given point, and they must define a right handed frame, $B = T \times N$. Consequently one can choose,

$$\begin{pmatrix} T(0) \\ N(0) \\ B(0) \end{pmatrix} = \begin{pmatrix} e_3 \\ e_1 \\ e_2 \end{pmatrix} = \begin{pmatrix} e_T \\ e_N \\ e_B \end{pmatrix}$$

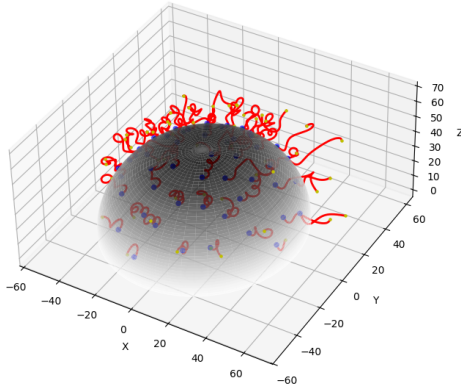
This last part I did not realize until long after implementing the original coefficient solver, which influenced the first few attempts. An illustration of those first few attempts can be seen below, with curvature and torsion functions that vary sinusoidally around κ_{init} and τ_{init} .



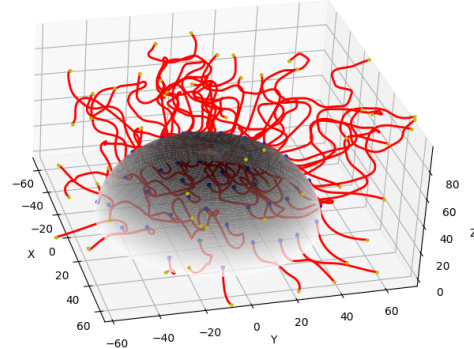
(a) First Attempt



(b) Length function w.r.t z



(c) High curvature values



(d) Lower curvature with fixed high length

3.2 A quick detour on Involutes

At this point I would like to take a short detour to explain the concept of an involute.

For a curve $\alpha : I \rightarrow \mathbb{R}^2$ or $\alpha : I \rightarrow \mathbb{R}^3$ p.b.a.l, an osculating circle is defined as the best circle approaching the curve at a point $\alpha(s_0)$, centered at,

$$\alpha(s_0) + \frac{1}{\kappa(s_0)}N(s_0)$$

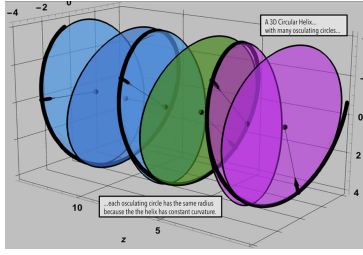
The curvature center. The evolute of the curve is defined as the trace of all the curvature centres,

$$\gamma(s) = \alpha(s) + \frac{1}{\kappa(s)}N(s)$$

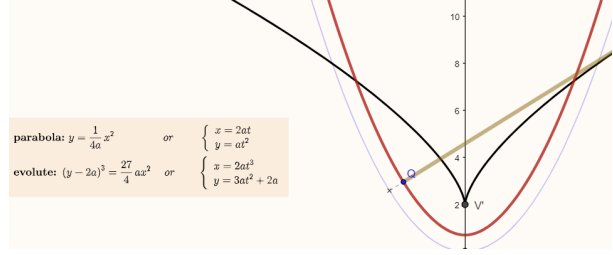
The involute is the geometric inverse of the evolute, an example of which is below,

$$I(s) = \alpha(s) + sN(s)$$

(source A) uses the the following analogy; if you have a taut string wrapped around the evolute, hold one end and unwind it whilst taut, the tip of the string traces out a new curve, the involute. Or another way would be that you start on the evolute, and then walk backwards along a specific direction for a distance equal to the length of string unwound.



(a) Osculating Circle(s)



(b) Involute Example

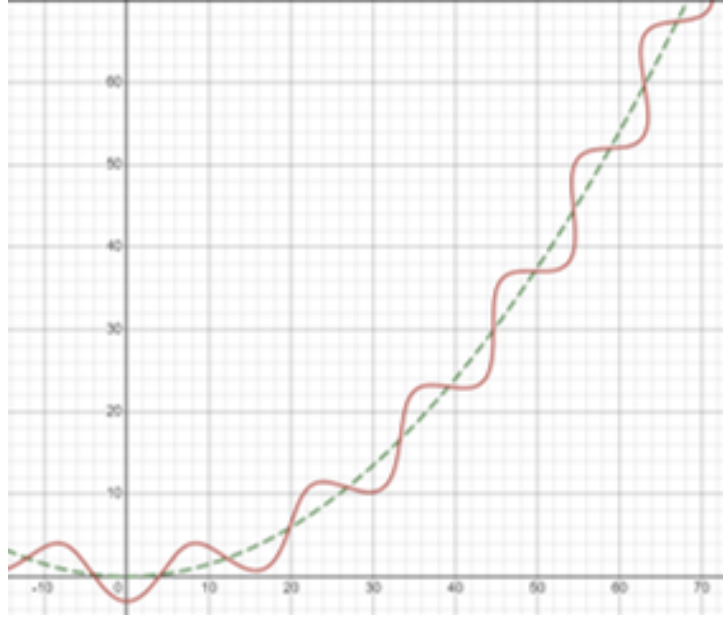


Figure 9: Parabola with Sinusoidal unwinding

In the figure above, you can see a function where this is applied parametrically to a parabola with a unique twist, which I set about deriving. After much trial and error, the derivation is as follows;
Suppose we have a parabola,

$$y = ax^2 \quad a \in \mathbb{R}$$

We can express this parametrically as,

$$\begin{aligned} \alpha(t) &= (t, at^2), \text{ so} \\ \alpha'(t) &= T_\alpha(t) = (1, 2at) \\ N_\alpha(t) &= (-2at, 1) \\ \|T_\alpha(t)\| &= \sqrt{1 + 4a^2t^2} \end{aligned}$$

Therefore the unit vectors are,

$$T_\alpha(t)^u = \frac{1}{\sqrt{1 + 4a^2t^2}}(1, 2at), \text{ and } N_\alpha(t)^u = \frac{1}{\sqrt{1 + 4a^2t^2}}(-2at, 1)$$

The arc length is defined by,

$$\begin{aligned}
 S(t) &= \int_0^t \|T_\alpha(u)\| du \\
 &= \int_0^t \sqrt{1 + 4a^2u^2} du \\
 &= \frac{t}{2} \sqrt{1 + 4a^2t^2} + \frac{1}{4a} \operatorname{arsinh}(2at) \\
 &\approx \frac{t}{2} \sqrt{1 + 4a^2t^2}, \quad \text{for small } a
 \end{aligned}$$

if we define a scalar function,

$$\lambda(t) = \cos(mt\sqrt{1 + 4a^2t^2}), \quad m \in \mathbb{R}$$

Then we can define another function,

$$\begin{aligned}
 P(t) &= \alpha(t) - \lambda(t)N_\alpha(t)^u \\
 &= \left(t + \frac{2at \cos(mt\sqrt{1 + 4a^2t^2})}{\sqrt{1 + 4a^2t^2}}, at^2 - \frac{\cos(mt\sqrt{1 + 4a^2t^2})}{\sqrt{1 + 4a^2t^2}} \right)
 \end{aligned}$$

Which is precisely the sinusoidal curve in the figure. In other words, if we treat the parabola as an evolute, this is its involute, but unlike the classic one, the scalar parameter varies sinusoidally w.r.t distance along the line; it dances along the way. This phenomena became incredibly useful in my application to the parting line.

3.3 Parting Line

Now we come to the main focus of the project, the parting line. In reality, this line is a curve that lies on the hemisphere, specifically part of a great circle. We can define a function $\lambda : I \rightarrow \mathbb{R}^3$,

$$\lambda(s) = P_0 \cos(\beta s) + VR \sin(\beta s) \quad P_0, V \in \mathbb{R}^3 \quad s \in [0, 1]$$

Where P_0 is the starting point, a unit vector V providing the initial direction, R the radius of the hemisphere and β a given frequency. A basic parting line is one defined by $P_0 = (50, 0, 0), V = (0, 0, 1), \beta = \pi$.

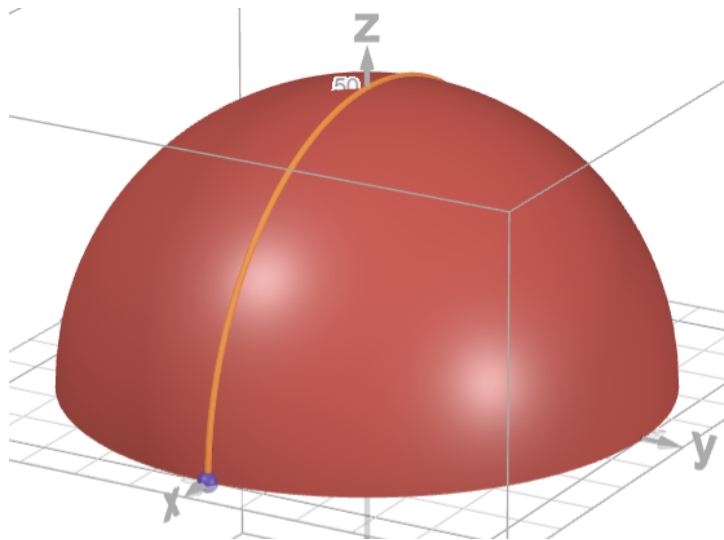


Figure 10: Basic Parting Line

Now, my idea for implementing the parting line was to rotate the basis of each tangent plane by an angle θ_{full} , s.t. the direction vector corresponding to the distance between a root and the nearest point on the parting line falls somewhere between the two vectors that span the plane.

We can define a squared distance function from the line to the root point P , such that

$$\begin{aligned} f(s) &= \|P - \lambda(s)\|^2 \\ &= \langle P - \lambda(s), P - \lambda(s) \rangle \\ f'(s) &= -2\langle T_\lambda(s), P - \lambda(s) \rangle \end{aligned}$$

$$\langle T_\lambda(s), P - \lambda(s) \rangle = 0$$

is the necessary condition. However, λ lies on the hemisphere; at any point $T_\lambda(s)$ lies in the tangent plane, therefore,

$$\langle T_\lambda(s), \lambda(s) \rangle = 0$$

As such the condition above can be simplified to,

$$\langle T_\lambda(s), P \rangle = 0$$

defining another great circle; for the basic parting line there is a single solution s^* on the hemisphere that minimizes the distance. The direction vector then is,

$$b = P - \lambda(s^*)$$

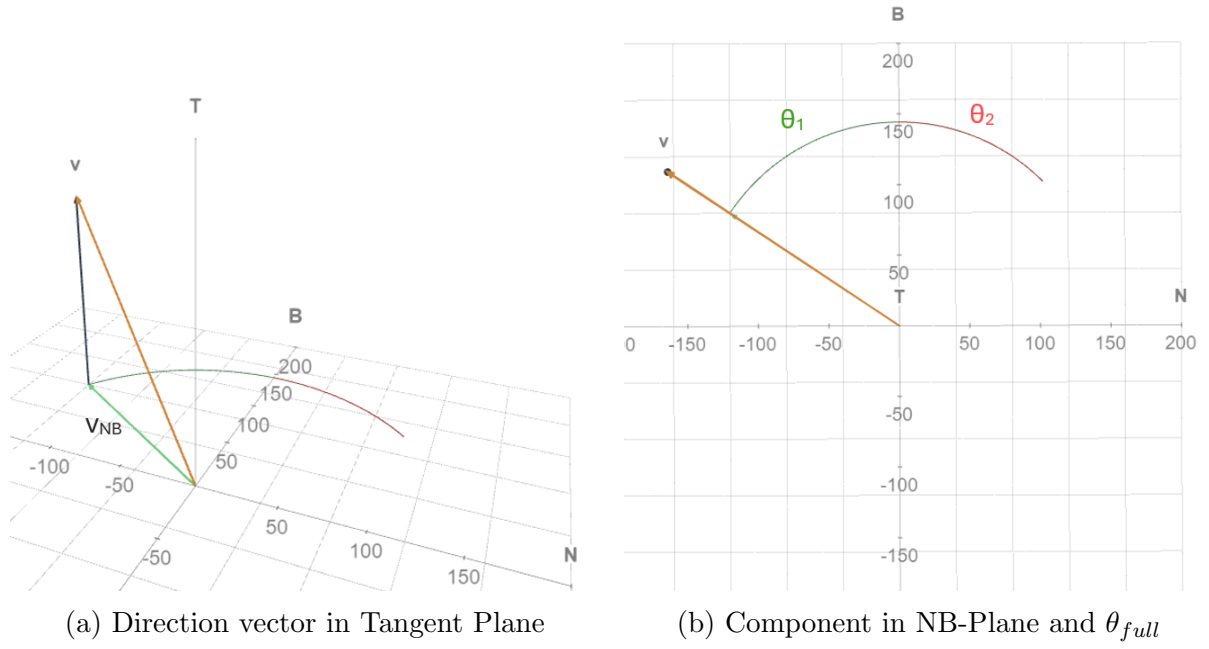
Expressing this vector in the tangent plane's basis,

$$v = (\hat{b} \cdot e_T)e_T + (\hat{b} \cdot e_N)e_N + (\hat{b} \cdot e_B)e_B = b_1e_T + b_2e_N + b_3e_B$$

An anticlockwise rotation of the bases about the normal axis, the tangent vector, can be expressed with the following matrix,

$$\begin{pmatrix} e'_T \\ e'_N \\ e'_B \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_{full} & \sin \theta_{full} \\ 0 & -\sin \theta_{full} & \cos \theta_{full} \end{pmatrix} \begin{pmatrix} e_T \\ e_N \\ e_B \end{pmatrix}$$

Let $\theta_{full} = \theta_1 + \theta_2$, where θ_1 is the angle between the component of v in the tangent plane and the binormal vector e_B . I am rotating the **basis**; θ_2 determines where the normal and binormal vectors, e_N, e_B , end up around v .



$$v_{NB} = v - b_1 e_T \quad (4)$$

$$= b_2 e_N + b_3 e_B \quad (5)$$

The signed angle from a vector u to a vector v for a plane in $\mathbb{R}^3$ is defined by the equation,

$$\theta = \text{atan2}(\langle u \times v, n \rangle, \langle u, v \rangle)$$

The two argument variation of the arctan function that preserves their signs to determine the correct quadrant. n is the normal vector to the plane determining which face we are on. Consequently,

$$\theta_1 = \text{atan2}(\langle e_B \times v_{NB}, e_T \rangle, \langle e_B, v_{NB} \rangle)$$

$$\theta_1 = \text{atan2}(\langle -b_2 e_T, e_T \rangle, b_3)$$

$$\theta_1 = \text{atan2}(-b_2, b_3)$$

Originally, I was planning on fixing $\theta_2 = \pi/2$, however this angle is what determines where the vector falls between those that span the plane; what If I made this vary across the scalp?

3.4 θ_2 function creation

I wanted this angle to vary so that when its close to 0, when the binormal vector is parallel to v and the shaft moves towards the right, we're at the back of the head. Then as we move towards the front, this angle increases, until the binormal is parallel to v at $\frac{\pi}{2}$. Initially this would have just been a simple straight line, however I wanted the rate at which it changed to slow down in the middle; to unwind as it dances along the way. Suppose we have a function $f : [-a, a] \rightarrow [c, d]$, $a > 0$, that is decreasing and defines a straight line, where $f(-a) = d$ and $f(a) = c$. We can express this function as,

$$f(x) = m_0(x + a) + d, \quad m_0 = \frac{c - d}{2a}$$

We can then define,

$$f_0(x) = \frac{1}{m_0}f(x) = x + a + \frac{d}{m_0}$$

and express this parametrically,

$$f_0(t) = \left(t, t + a + \frac{d}{m_0} \right), \quad t \in [-a, a]$$

then,

$$T(t) = (1, 1) \quad N(t) = (-1, 1), \quad \text{so,}$$

$$\|T(t)\| = \sqrt{2}, \quad S(t) = \sqrt{2}t \quad \text{and} \quad N(t)^u = \frac{1}{\sqrt{2}}(-1, 1)$$

We can define the scaling function,

$$\lambda(t) = A \sin(m\sqrt{2}t), \quad A, m \in \mathbb{R}$$

The involute of the evolute $f_0(t)$ is then,

$$\begin{aligned} f_0^{\sim}(t) &= f_0(t) + \lambda(t)N(t)^u \\ &= \left(t - \frac{\lambda(t)}{\sqrt{2}}, t + a + \frac{d}{m_0} + \frac{\lambda(t)}{\sqrt{2}} \right), \quad t \in [-a, a] \end{aligned}$$

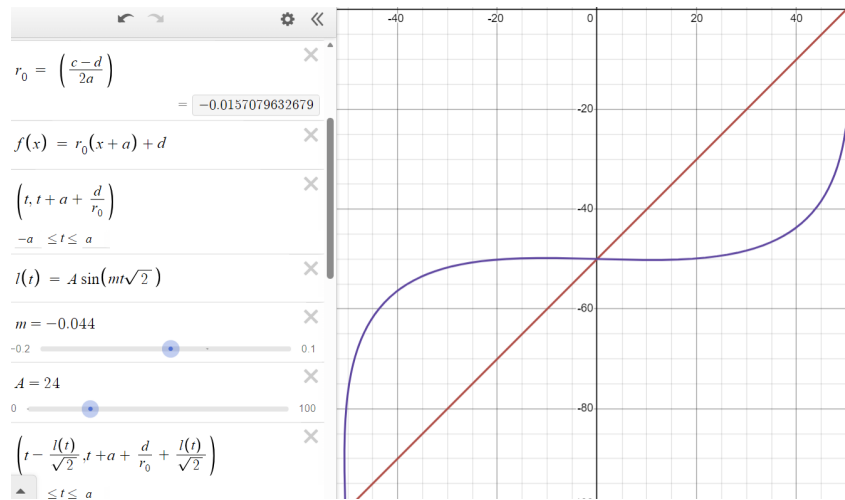
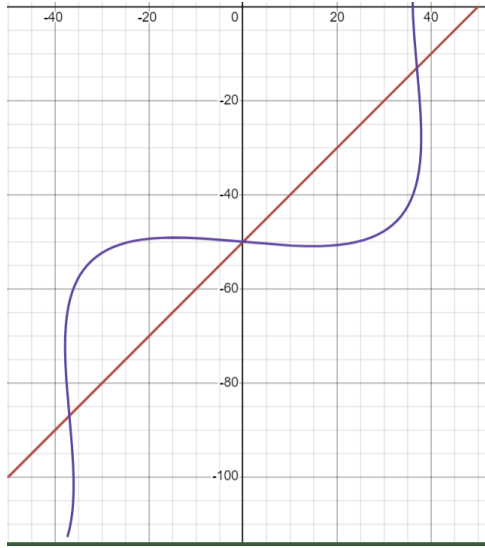
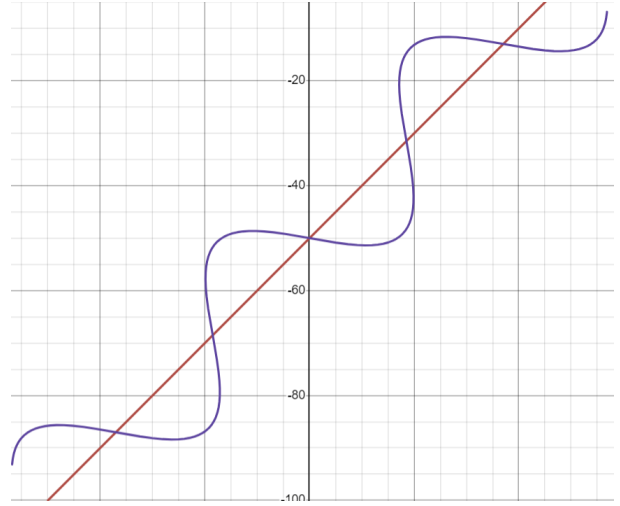


Figure 12: Goal function, $m \approx -0.044$, $A = 24$

(a) $m = -0.16, A = 20$ (b) $m = -0.12, A = 12$

m and A are the only variables from the scaling function, and I needed to fix them such that they satisfy a few conditions,

1. The functions must have equal values at their endpoint.
2. m must be negative, such that the function is initially concave down
3. m must correspond to a specified number of intersections.
4. A must be bounded, such that the transformed function never exceeds the domain.

Let N be the order of a function with $n_I = 2N - 1$ intersection points (excluding the endpoints). For the first condition, we need

$$f_0(t) = f_0(t)$$

, which implies

$$\lambda(t) = 0, \quad mt = \frac{n\pi}{\sqrt{2}}, \quad \forall n \in \mathbb{Z}$$

Take Figure 10, the goal function. This is a function of order $N = 1$, so $n_I = 1$. We can observe that,

$$\begin{aligned} t_0 &= -a \\ t_1 &= -a + \frac{a - (-a)}{2} = 0 \\ t_2 &= a + 2 \frac{a - (-a)}{2} = a \end{aligned}$$

From desmos, the value of m that corresponds with equal endpoints ≈ -0.04443 .

Plugging this into the equation derived from the condition we obtain,

$$\begin{aligned} n_0 &\approx 0.999... = 1 \\ n_1 &\approx 0.00... = 0 \\ n_2 &\approx -0.990... = -1 \end{aligned}$$

I then looked at $N = 2, n_I = 3$, and observed a similar pattern, with desmos giving $m \approx -0.0889$, and

$$\begin{array}{ll} t_0 = -a, & n_0 = 2 \\ t_1 = -a + \frac{a - (-a)}{4} = -\frac{a}{2}, & n_1 = 1 \\ t_2 = -a + 2\frac{a - (-a)}{4} = 0, & n_2 = 0 \\ t_3 = -a + 3\frac{a - (-a)}{4} = \frac{a}{2}, & n_3 = -1 \\ t_4 = -a + 4\frac{a - (-a)}{4} = a, & n_4 = -2 \end{array}$$

From this, we can deduce for $i \in [0, n_I + 1], \mathbf{Z}$,

$$\begin{aligned} t_i &= -a + 2\frac{ia}{n_I + 1} \quad (= -a + \frac{ia}{N}) \\ n_i &= \frac{1 + n_I}{2} - i \quad (= N - i) \end{aligned}$$

Therefore,

$$\begin{aligned} m &= \frac{\pi}{\sqrt{2}} \frac{n_i}{t_i} \\ &= -\frac{N\pi}{a\sqrt{2}}, \quad N \geq 1 \end{aligned}$$

Next consider the derivative,

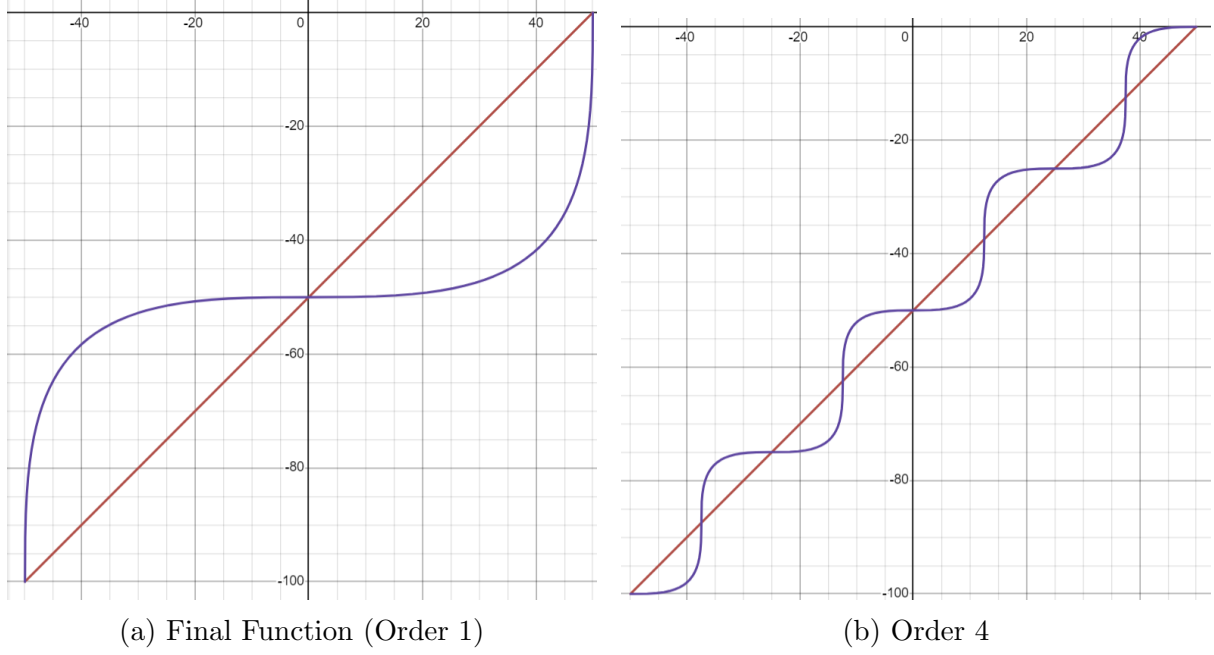
$$f_0(\tilde{t})' = \left(1 - \frac{\lambda'(t)}{\sqrt{2}}, 1 + \frac{\lambda'(t)}{\sqrt{2}}\right)$$

From the goal function, I observed that A should reach its upper bound when $f_0(\tilde{-a})'_x = 0$. If we fix $A > 0$, and

$$\begin{aligned} \lambda'(-a) &= \sqrt{2} \\ Am\sqrt{2}\cos(-am\sqrt{2}) &= \sqrt{2} \\ A &= \frac{1}{m\cos(-am\sqrt{2})}, \quad \text{at } t = -a, \end{aligned}$$

Then we can fix

$$A \in \left(0, \frac{1}{\|m\|}\right]$$



Finally, f needs to be a function of the x coordinate; I need to undo the parameterization. Recall that

$$f(x) = m_0 f_0(\tilde{x})$$

. Let,

$$f_0(t) = (h(t), g(t))$$

Where,

$$h(t) = t - \frac{\lambda(t)}{\sqrt{2}}, \quad g(t) = t + a + \frac{d}{m_0} + \frac{\lambda(t)}{\sqrt{2}}$$

Then,

$$x = h(t) \implies t = h^{-1}(x) \implies f(x) = m_0 g[h^{-1}(x)], \quad x \in [-a, a]$$

But for this to be a valid function, I need to prove that h is invertible; that it is strictly increasing/decreasing.

$$h(x) = x - \frac{\lambda(x)}{\sqrt{2}} = x - \frac{A}{\sqrt{2}} \sin(m\sqrt{2}x)$$

$$h'(x) = 1 - Am \cos(m\sqrt{2}x)$$

So h' oscillates between $1 - Am$ and $1 + Am$. We can rewrite these as $1 + A\|m\|$ and $1 - A\|m\|$. $1 + A\|m\| > 0$. If $1 - A\|m\| < 0$ then $A > \frac{1}{\|m\|}$, which contradicts the bound above. Therefore, $1 - A\|m\| \geq 0 \implies h'$ is non-decreasing. $h'(x) = 0$ by construction only occurs at a finite number of points, such as the left boundary.

For $h(x)$ to be strictly increasing, we need $x_1 < x_2 \implies h(x_1) < h(x_2)$.

$$\begin{aligned} h(x_2) - h(x_1) &= x_2 - \frac{A}{\sqrt{2}} \sin(m\sqrt{2}x_2) - x_1 + \frac{A}{\sqrt{2}} \sin(m\sqrt{2}x_1) \\ &= (x_2 - x_1) - \frac{A}{\sqrt{2}} [\sin(m\sqrt{2}x_2) - \sin(m\sqrt{2}x_1)] \end{aligned}$$

By the Mean Value Theorem, $\sin A - \sin B \geq -(A - B)$, $\implies$

$$\begin{aligned} h(x_2) - h(x_1) &\geq (x_2 - x_1) + \frac{A}{\sqrt{2}} (m\sqrt{2}x_2 - m\sqrt{2}x_1) \\ &\geq (x_2 - x_1) + Am(x_2 - x_1) \\ &\geq (1 + Am)(x_2 - x_1) \\ &\geq 0 \end{aligned}$$

$h(x)$ is a strictly increasing function, invertible. My θ_2 function can be therefore be summarized as follows:

Let $\theta_2 : [-R, R] \rightarrow [0, \pi/2]$ be a function such that

$$\theta_2(x) = m_0 g[h^{-1}(x)], \quad \text{where}$$

$$m_0 = -\frac{\pi}{2R}$$

$$h(x) = x - \frac{\lambda(x)}{\sqrt{2}}, \quad g(x) = x + R + \frac{\pi}{2m_0} + \frac{\lambda(x)}{\sqrt{2}}$$

$$\lambda(x) = A \sin(m\sqrt{2}x), \quad m = -\frac{\pi}{R\sqrt{2}}, \quad A \in \left(0, \frac{1}{\|m\|}\right]$$

Consequently,

$$\theta_{full}(x) = \text{atan2}(-b_2, b_3) + m_0 g[h^{-1}(x)]$$

where x is the first coordinate of that guide root. For the other roots, θ_1 was computed to ensure initial direction, and then a weighted average of θ_2 added on with the same pareto distributed weights previously applied to the curvature/torsion vector.

Regarding the parting line, the Distribution was shifted so that the other hairs only considered the guide hairs on the same side as them, to avoid incorrect rotation; when not using the parting line, θ_{full} is a random angle taken from the interval $[-2\pi, 2\pi]$, and similarly distributed, however it was not the main focus. Furthermore, from the above we can deduce that,

$$\theta_{full}(-x) = 2 \text{atan2}(-b_2, b_3) - \theta_{full}(x)$$

Therefore the function is not symmetric; recall that $T'_x = \kappa N_x$, and N_x is determined by this rotation. This led to systematic drift on either side in different directions. At the time of completing this, this issue was resolved cosmetically by mirroring each curve across the parting line.

3.5 Old Rotation attempts and subsequent Parting Adjustment

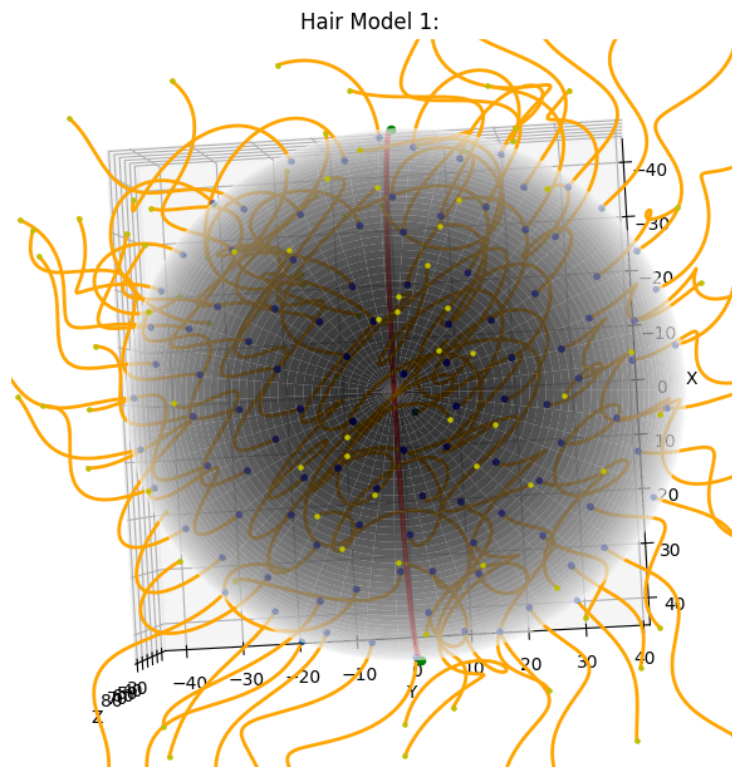


Figure 15: Basic Parting, $\kappa_0 \in [0.0645, 0.09]$, $\tau_0 \in [0.11, 0.15]$

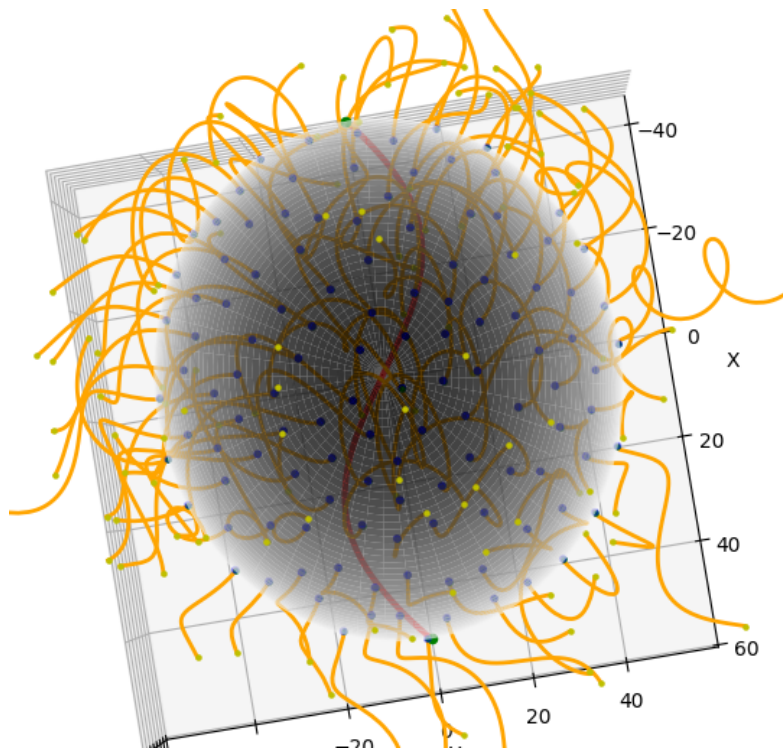


Figure 16: Old Sin Parting, $\kappa_0 \in [0.0645, 0.09]$, $\tau_0 \in [0.11, 0.15]$

You will notice that while the rotation occurs somewhat, it does not do so perfectly. Moreover, the hairs are crossing over the parting line, which defeats its purpose. This is largely because I had not yet properly vectorized the rotation distribution yet. I will introduce the adjustments made during the solution generation. Recall the position is determined by the equation,

$$\alpha'(s) = T(s)$$

While T is set normal to the tangent plane, this equation alone cannot influence the behaviour of the hair strand further down its arc length. I first let the solution iterate normally for N steps, $N \ll n$, necessary to gain some initial height. For all $n > N$, I needed a way to ensure the hair's actual position would drop at each time step. At any point $M = (x_m, y_m, z_m)$ on the curve, we can define two values,

$$G = z_m - \sqrt{R^2 - x_m^2 - y_m^2} \quad \text{and} \quad S = \frac{C}{n_0} \sum_m |R - z_m|$$

where,

$$C = 0.3e^{-0.6\kappa_{ave}}$$

Despite its seeming arbitrariness, this is one of the most important values in the entire setup, as it controls rate of decay after leaving the hemisphere. It is a decreasing function of κ_{ave} ; a cosmetic change reflecting the decreased influence of the gravitational force. The constants are currently designed for a κ_p value of 1. Then,

$$z_{new} = z_m - \mathbf{w}_m G \quad \text{for points above the hemisphere, and} \quad (6)$$

$$z_{new} = z_m - \mathbf{w}_m S \quad \text{after it leaves} \quad (7)$$

Where $\mathbf{w}$ is a vector of pareto distributed weights; recall the one applied previously, but instead of distance, these were a function of the assigned curvature values themselves, so

$$f_K(\kappa_{init}) = \frac{\alpha(\kappa_{min})^\alpha}{\kappa_{init}^{\alpha+1}} = v_i^*, \quad \text{however} \quad v_i = \frac{n_0}{500} \frac{v_i^*}{\sum_i v_i^*}$$

v_i^* is decreasing in n_0 . The factor of $\frac{n_0}{500}$ ensures the visual length remains constant across n_0 . The minimum curvature is something which will be discussed more in the following section, as it may not be the bottom of the interval. I needed to take a closer look at curvature across the strand itself and its relationship to length.

4 A closer look at Curvature and Length

4.1 Curvature consideration

Recall that the hairs can be likened to independent springs; as the shaft grows, gravity exerts milligrams of force. Axial elongation of the hair shaft is a consequence of this gravitational force, but recall how around the perception interval this shifts to torsional buckling. Whilst a proper physical consideration of hair is left for Phase 1, I wanted to design a curvature function that reflected these phenomena.

For hair with 0 curvature, perceived length (L_p), = actual length (L_r). As curvature increases, perceived length, as a function of actual length and curvature, decreases. Curvature may also decrease due to tension and damage, but these are physical properties not considered in Phase 0. Curvature is mathematically defined as $1/R$, where R is the radius of the osculating plane at a point on the shaft. There's a gravitational force associated with each point on the strand; suppose a $G(z)$, linearly increasing. As we move along the shaft curvature decreases due to G , generating that axial elongation. I chose to model the curvature κ then, as a decreasing function of G , the distance s along the curve, and the initial curvature value κ_{init} . The model then is conscious of the perceived length, dependent on curvature, and not a variable actual length. Let,

$$\kappa = \kappa(G, \kappa_{init}, s), \quad \text{where}$$

$$\kappa(G, \kappa_{init}, 0) = \kappa_{init}, \quad \kappa(G, \kappa_{init}, L_r) = \kappa_{LB}; \quad \kappa'_s \leq 0 \quad \forall s \geq 0$$

Where $\kappa_{LB} = p\kappa_{init}$ is the lower bound of the curvature function, the lowest value it can decrease to along the strand. Recall that κ_{init} is taken from the curvature interval, $[\kappa_1, \kappa_2]$. I wanted the percentage difference between κ_{init} and its lower bound to be equal to that between the interval boundaries. As such,

$$p = \frac{\kappa_1}{\kappa_2}$$

As κ_{init} increases, the rate at which κ changes along the shaft decreases until practically 0 at some specified κ value; this is the previously mentioned perception value, κ_p , after which torsion takes over as the main cause of hair variation. As such, the following conditions must be satisfied;

1. $\frac{\partial \kappa'_s}{\partial \kappa_{init}} > 0$ (less negative as it increases)
2. $\kappa'_s(G, \kappa_{init}, s) = 0$ for $\kappa_{init} \geq \kappa_p$
3. $\lim_{\kappa_{init} \rightarrow \kappa_p} \kappa'_s(G, \kappa_{init}, s) = 0$

The preliminary function was then,

$$\kappa(G, \kappa_{init}, s) = \kappa_{init} + m(\kappa_{init}) \frac{s}{\phi(G)} \quad \text{for } 0 < \kappa_{init} < \kappa_p$$

Where the function m , being the slope, satisfies the conditions given above. I chose,

$$m(\kappa_{init}) = -(\kappa_p - \kappa_{init})^2$$

4.2 Length consideration

As for $\phi(G)$, recall the initial condition,

$$\kappa(G, \kappa_{init}, L_r) = \kappa_{LB}, \quad \kappa_{LB} = p\kappa_{init}$$

If we plug this into the preliminary function and rearrange, we obtain,

$$L_r(G) = \frac{\phi(G)(\kappa_{LB} - \kappa_{init})}{m(\kappa_{init})} = \frac{\phi(G)\kappa_{init}(p - 1)}{m(\kappa_{init})}$$

You can consider $L_p = L_r(G)$ as precisely the perceived length, one that varies depending upon the general curvature of as well as the weight associated with the shaft.

On the other hand,

$$\frac{\partial \kappa}{\partial G} = \frac{-m(\kappa_{init})\phi'(G)s}{\phi(G)^2} \implies \frac{\partial \kappa}{\partial G} < 0 \quad \text{iff} \quad \phi'(G) < 0$$

Moreover, I wanted this rate of change to be faster with respect to G than κ_{init} . Let $c_0 \in \mathbb{R}_{>0}$, s.t.

$$\begin{aligned} \phi(G(R)) = c_0 p^2 R &\implies c_0 p R e^{-bG(R)} = c_0 p^2 R \implies G(R) = \frac{\ln p}{-b} > 0 \\ \phi(G(0)) = c_0 p R &\implies c_0 p R e^{-bG(0)} = c_0 p R \implies G(0) = 0 \end{aligned}$$

We can construct a function,

$$G(z) = \frac{-\ln p}{bR} z \quad \text{s.t.} \quad \phi(G(z)) = c_0 R p^{1+\frac{z}{R}}, \quad z \in [0, R]$$

Boundaries needed to be observed to prevent the length from blowing up as κ_{init} approaches κ_p , so I capped the curvature values at $q\kappa_p$, $q \in (\frac{1}{2}, 1)$. Likewise for very small curvatures the equation became more stiff or broke down entirely; as such, I fixed the lower bound at $(1 - q)\kappa_p$. Next, let,

$$c_0 = \begin{cases} \frac{(17q - 4)\kappa_p - 9\kappa_{ave}}{2q - 1} & (1 - q)\kappa_p < \kappa_{ave} \leq q\kappa_p \\ 6k_p & \text{otherwise} \end{cases}$$

Where κ_{ave} is the average curvature for a given curvature interval, $[\kappa_1, \kappa_2]$. c_0 ensures there is a perceivable length across the scalp, the κ_p coefficient determines the max perceivable length, the κ_{ave} coefficient the rate at which it changes over for different intervals. The final functions then were,

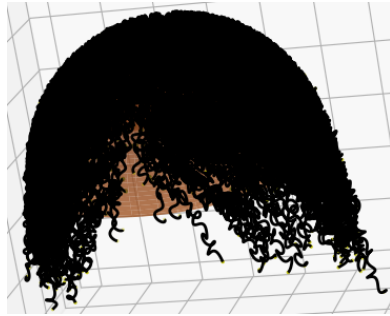
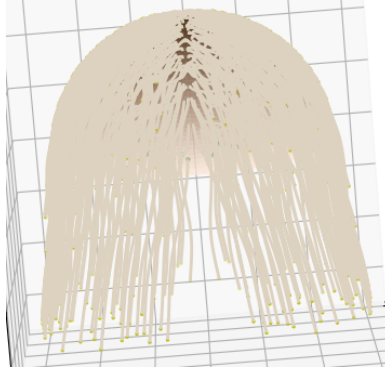
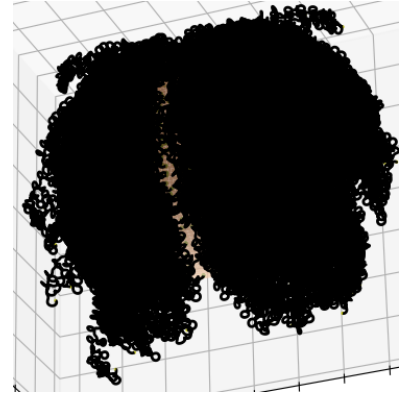
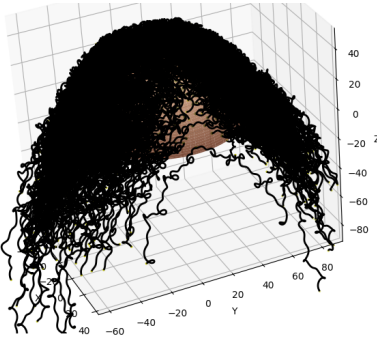
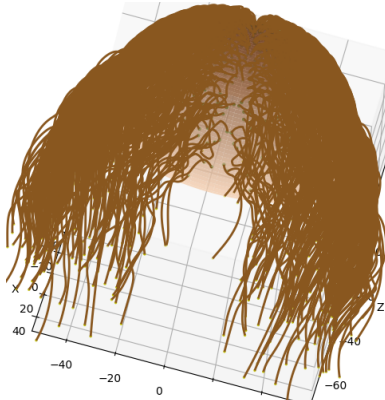
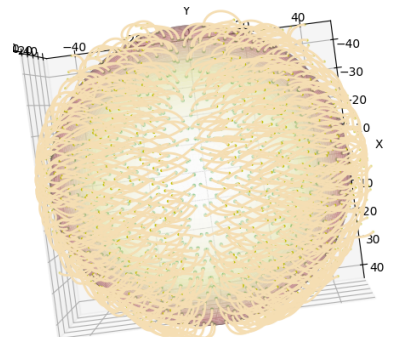
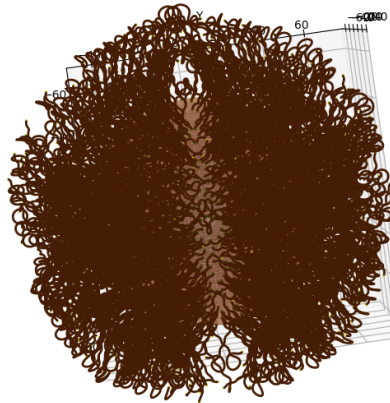
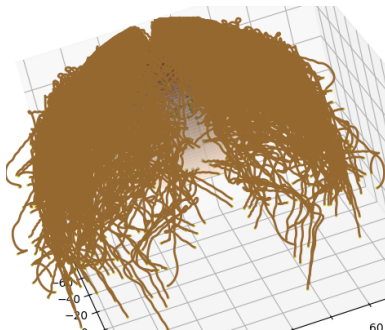
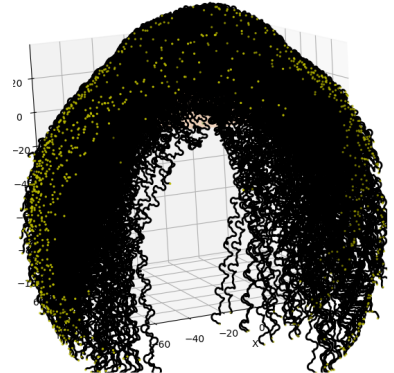
$$L_r(G) = \frac{\phi(G)\kappa_{init}(p - 1)}{m(\kappa_{init})}, \quad (1 - q)\kappa_p \leq \kappa_{init} \leq q\kappa_p$$

$$\phi(G(z)) = c_0 R p^{1+\frac{z}{R}}, \quad z \in [0, R], \quad m(\kappa_{init}) = -(\kappa_p - \kappa_{init})^2,$$

$$\kappa(G, \kappa_{init}, s) = \kappa_{init} + m(\kappa_{init}) \frac{s}{\phi(G)}, \quad s \in [0, L_r(G)]$$

4.3 Final Showcase

Value assignment takes half a second; the first N steps above the scalp takes 2s at its best, half the runtime. Where curvature and torsion match, the perceivable length is generally longer; where they don't, especially with large torsion, there is more lateral movement, therefore a shorter length. For large torsion gaps the model will break down, particularly towards the higher values if its greater than the curvature. The hairs will overwhelm the parting line, such as third row (c), or the model will break down. The below have 1000 strands from $n_0 = 1500$, 300 guide hairs, α parameters 1.2 and 1.4.

(a) $\kappa[0.62, 0.69], \tau[0.12, 0.17]$ (b) $\kappa[0.26, 0.28], \tau[0.12, 0.17]$ (c) $\kappa[0.63, 0.77], \tau[0.63, 0.77]$ (a) $\kappa[0.42, 0.57], \tau[0.32, 0.35]$ (b) $\kappa[0.29, 0.36], \tau[0.2, 0.28]$ (c) $\kappa[0.12, 0.26], \tau[0.07, 0.13]$ (a) $\kappa[0.34, 0.45], \tau[0.42, 0.46]$ (b) $\kappa[0.34, 0.45], \tau[0.72, 0.76]$
High τ Large Gap Break(c) $\kappa[0.83, 0.9], \tau[0.83, 0.9]$

5 Afterthoughts

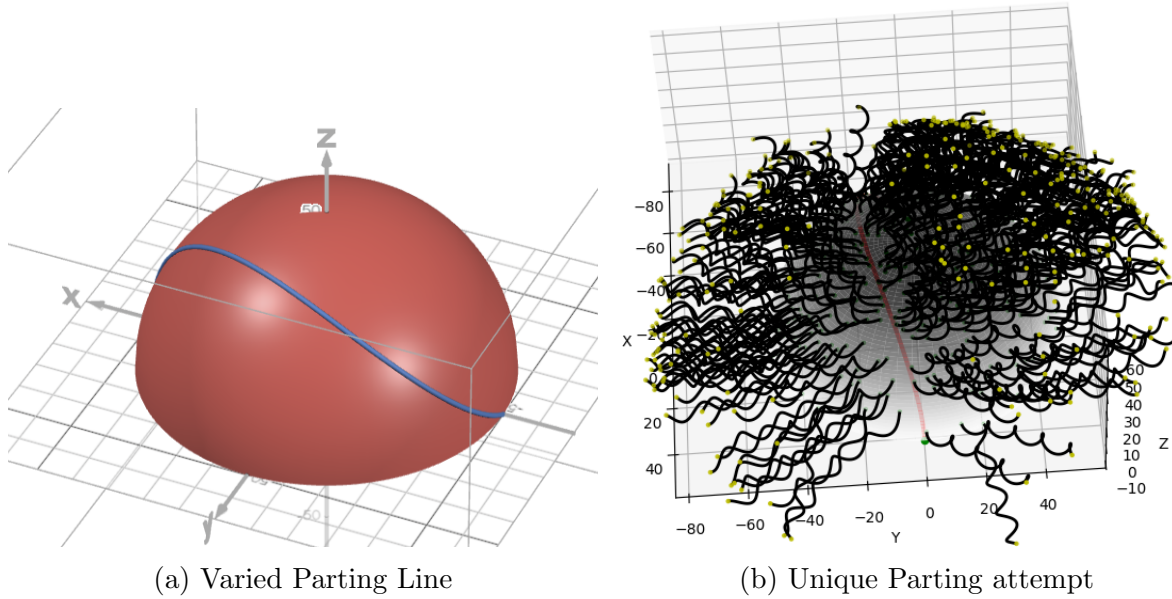
The next steps in this project will likely involve the following:

1. Refining this setup, there remain a few arbitrary choices due to various constraints.
2. Modelling the cross-sectional geometries of the hair strands, treating them as proper objects on top of regular curves, something akin to a Phase 0.5.
3. The curvature map across the surface is essential to the layout and length of the hair; it is kept constant for this showcase, but can be explored further for specific variation across the scalp.
4. Investigating the physical characteristics of hair, elasticity and other parameters as well and incorporating them into the model, this would be in Phase 1.
5. Using these to take a preliminary peak into the torsion domain. In addition to this, there are countless other interesting applications. Recall that this was merely a basic parting line. We can also consider the following,

$$\lambda(s) = P_0 \cos(\beta s) + V(s)R \sin(\beta s) \quad P_0 \in \mathbb{R}^3 \quad s \in [0, 1]$$

$$V(s) = (0, \sin(A\beta s), \cos(A\beta s)) \quad A \in \mathbb{R}$$

The z coordinate must remain positive for all s for λ to remain on the hemisphere, as such $|A|$ is bounded above by $\frac{\pi}{2\beta}$, which in the case below is $\frac{1}{2}$,



However I have yet to put together a proper solution for mirroring hairs across this parting line in a way I like, which is only an issue due to the drift problem created by the θ_{full} asymmetry.

As of 09/2026, I am approaching the 3rd year of my integrated masters studying mathematics at Durham University within the United Kingdom. I am looking for internships with companies and institutes in the fields of cosmetic science, Hair R&D and Modelling, alongside sponsorship for my degree and beyond.

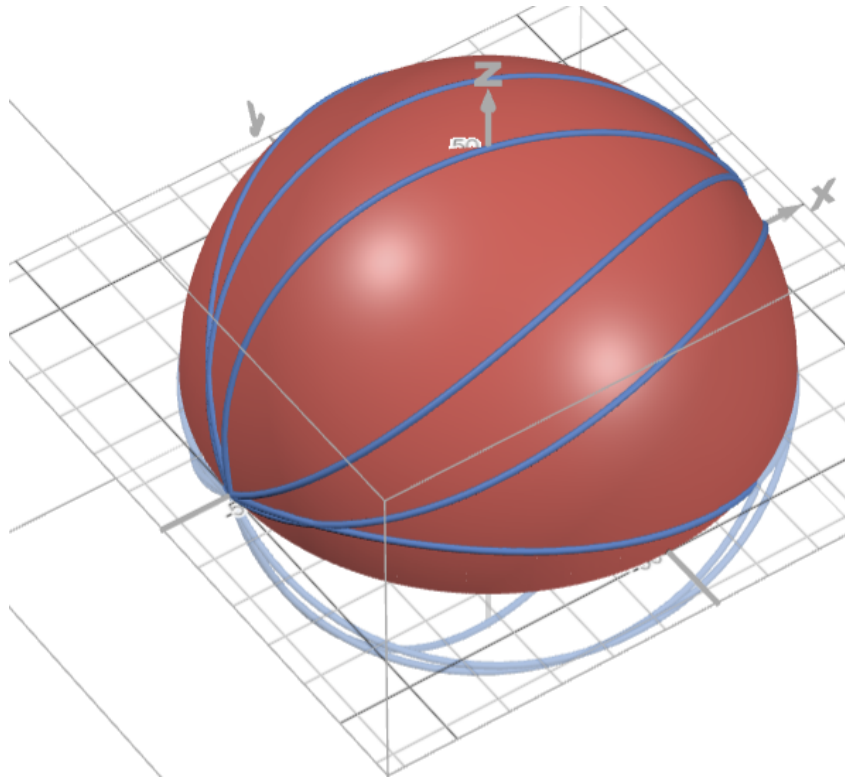


Figure 21: Future Parting applications

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